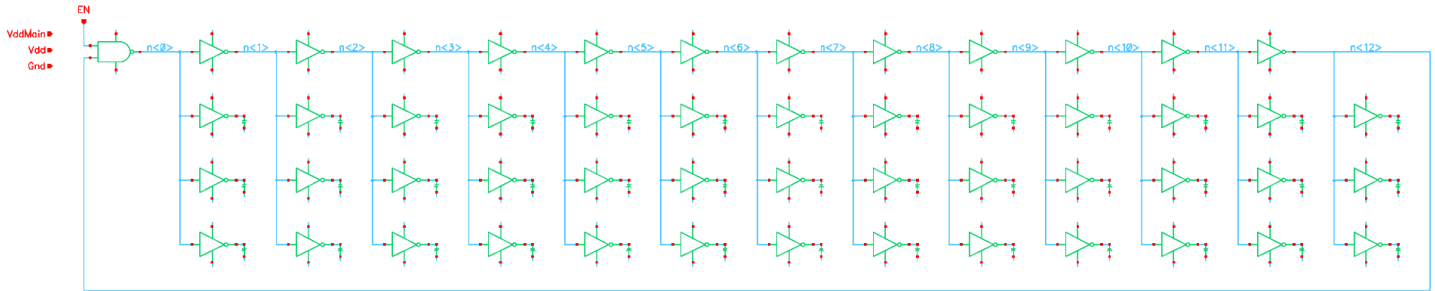


HW2 Virtuoso Repo: <https://github.com/vishalkevat007/Advanced-VLSI-Design-ECE695-Spring2026-Homeworks.git>

In this problem, we will find the VDD at which the total energy consumption of a circuit is minimized. For that we will use a ring oscillator circuit.

(a) Implement the schematic of 13 stage ring oscillator with fanout of 4 (FO4) based on CMOS inverters and a NAND gate as shown below.



(b) Size the transistors of the inverter and NAND gate to have almost equal rise and fall times. All the inverters should be sized identically. Clearly show what sizes have been used and how you determined those sizes.

Solution: For Inverter sizing:

Set width of N-MOS as minimum width permitted by the technology. Therefore, width_n = 120 nm

To have almost equal rise and fall times at the output for a load capacitance of 2fF of the inverter, I did pulse input with Vdd 1.1 V and swept width of PMOS to get equal rise and fall time values as shown by below simulation.

Name	Type	Details	Value
	signal	/In	
	signal	/Out	
tphl_in_falls_Out_rises	expr	delay(?wf1 VT("/In") ?value1 0.55 ?edge1 "falling" ?nth1 1 ?td1 0.0 ?to...	29.42p
tphl_in_rises_Out_falls	expr	delay(?wf1 VT("/In") ?value1 0.55 ?edge1 "rising" ?nth1 1 ?td1 0.0 ?tol...	30.12p
rise_time	expr	riseTime(VT("/Out") 0 nil 1.1 nil 10 90 nil "time")	42.24p
fall_time	expr	fallTime(VT("/Out") 1.1 nil 0 nil 10 90 nil "time")	42.17p

We get width_p = 180 nm

For NAND sizing:

NAND gate was sized according to above reference inverter through resistance equalization. Therefore Pull-up PMOS width = 180 nm, Pull-down stack NMOS width = 240 nm

The simulation below shows that we get equal rise and fall times. This confirms our sizing approach for NAND.

Name	Type	Details	Value
A_input	signal	/A	
B_input	signal	/B	
Output	signal	/Out	
tphl_A_rises_Out_falls	expr	delay(?wf1 VT("/A") ?value1 0.55 ?edge1 "rising" ?nth1 1 ?td1 0.0 ?tol1...	31.49p
tphl_A_falls_Out_rises	expr	delay(?wf1 VT("/A") ?value1 0.55 ?edge1 "falling" ?nth1 1 ?td1 0.0 ?tol...	31.35p
rise_time	expr	riseTime(VT("/Out") 0 nil 1.1 nil 10 90 nil "time")	45.7p
fall_time	expr	fallTime(VT("/Out") 1.1 nil 0 nil 10 90 nil "time")	45.71p

(c) Perform transient simulations to determine the (i) frequency, (ii) total power and (iii) leakage power of the ring oscillator for VDD ranging from deep sub-threshold to super threshold regions. You will need to determine the range depending on the technology that you use. Make sure the total power and leakage power is only for the ring oscillator comprising of 12 inverters and the NAND gate. The power expended by the load inverters should not be included in your calculations. For that, consider using one power supply for the gates in the ring (12 inverters + NAND) and a second supply for the load inverters (not in the ring).

(Hint: Leakage power can be obtained by setting EN = 0 which suppresses oscillations in the circuit. Total power can be obtained by setting EN=1 which starts oscillations. Note, power = VDD * average current drawn from VDD)

Solution:

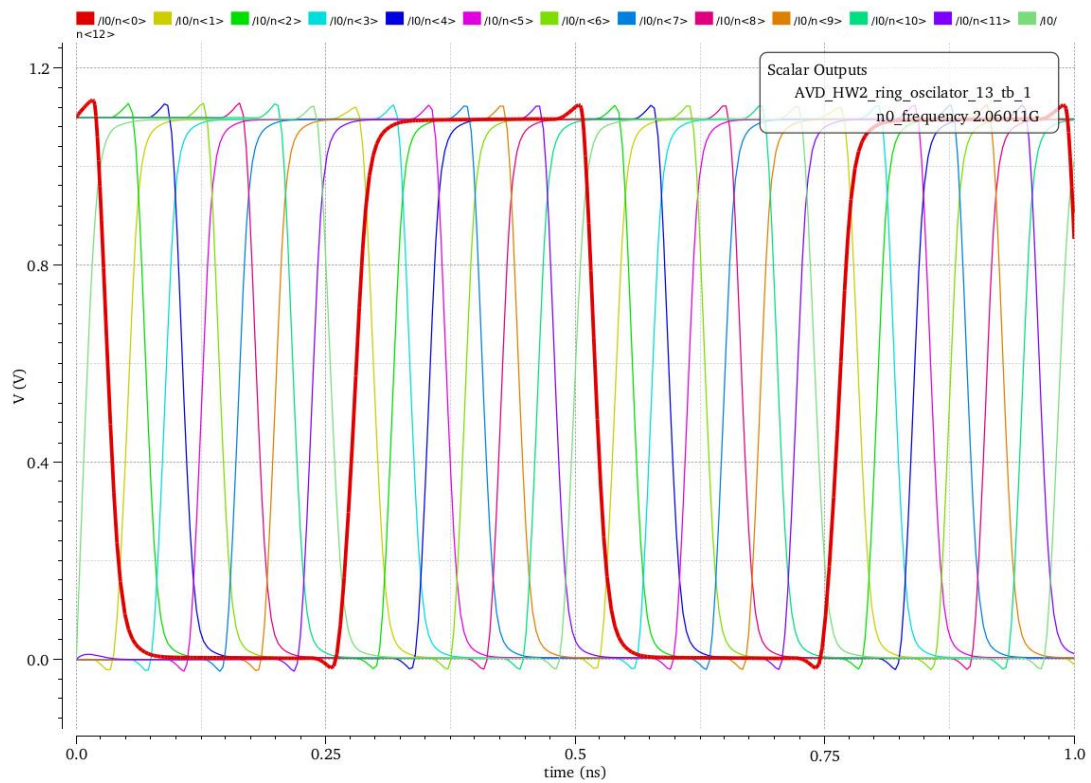
The below results are from simulation done for VDD ranging from 0.1 to 1.2 V

VDD (V)	I_{avg} (A)	P_{tot} (W)	P_{leak} (W)	Timeperiod T (s)	Frequency f (Hz)
0.10	20.34p	2.034p	1.344p	162.5u	6.156K
0.15	54.75p	8.212p	2.572p	48.73u	20.52K
0.20	206.8p	41.36p	4.303p	13.73u	72.83K
0.25	872p	218p	6.627p	3.81u	262.4K
0.30	3.57n	1.071n	9.648p	1.096u	912.6K
0.35	13.17n	4.611n	13.5p	344.2n	2.905M
0.40	42.12n	16.85n	18.32p	122.9n	8.14M
0.45	116.7n	52.51n	24.32p	50.08n	19.97M
0.50	288.9n	144.4n	31.7p	22.72n	44.01M
0.55	656.1n	360.8n	40.7p	11.18n	89.43M
0.60	1.378u	826.8n	51.63p	5.949n	168.1M
0.65	2.632u	1.711u	64.86p	3.461n	288.9M
0.70	4.533u	3.173u	80.88p	2.218n	450.9M
0.75	7.095u	5.321u	100.2p	1.553n	643.7M
0.80	10.24u	8.195u	123.7p	1.171n	853.7M
0.85	13.88u	11.8u	151.9p	935.4p	1.069G
0.90	17.93u	16.14u	185.9p	778.8p	1.284G
0.95	22.3u	21.19u	226.9p	670.3p	1.492G
1.00	26.93u	26.93u	276.2p	591.6p	1.69G
1.05	31.8u	33.39u	335.4p	532p	1.88G
1.10	36.86u	40.54u	406.6p	485.7p	2.059G
1.15	42.08u	48.39u	492.1p	448.7p	2.229G
1.20	47.46u	56.95u	595p	418.6p	2.389G

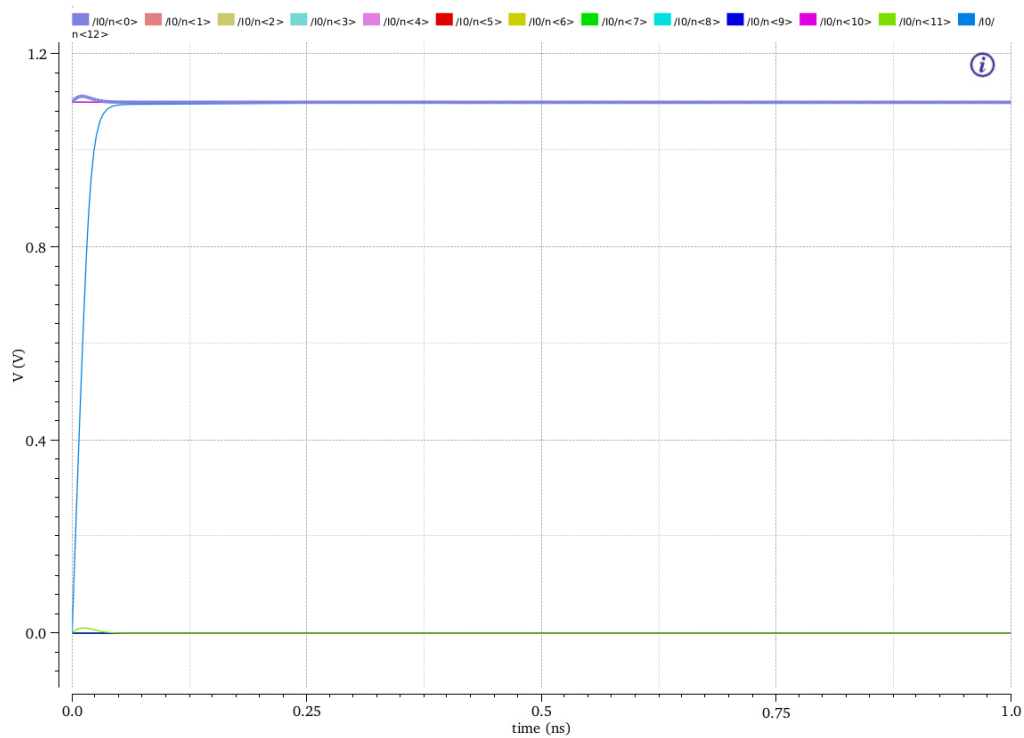
(d) Include the waveforms in your submission clearly showing the oscillations when EN=1 and no oscillations when EN=0.

Solution:

Ring Oscillations when EN=1



Ring Oscillations when EN=0



Process your results in the following manner

(A) Obtain dynamic power = Total power – Leakage power. Since this is based on inverter based ring oscillator with EN always ON, the switching activity is 100%.

(B) In a typical circuit, switching activity is low. In this assignment, we will use a switching activity of 1% to obtain dynamic power (PDYN) corrected for the probability of switching by multiplying the dynamic power obtained in (A) by 0.01. This is equivalent to saying EN is on for 1% of the total time of our interest.

(C) Multiply the dynamic power by the time period of the ring oscillator. This gives us the dynamic energy per cycle.

(D) Multiply the leakage power by the time period of the ring oscillator. This gives us the leakage energy per cycle.

(E) Add the results of (C) and (D) to obtain total energy per cycle.

Solution:

Here is the table for (A) to (E) for all VDD points:

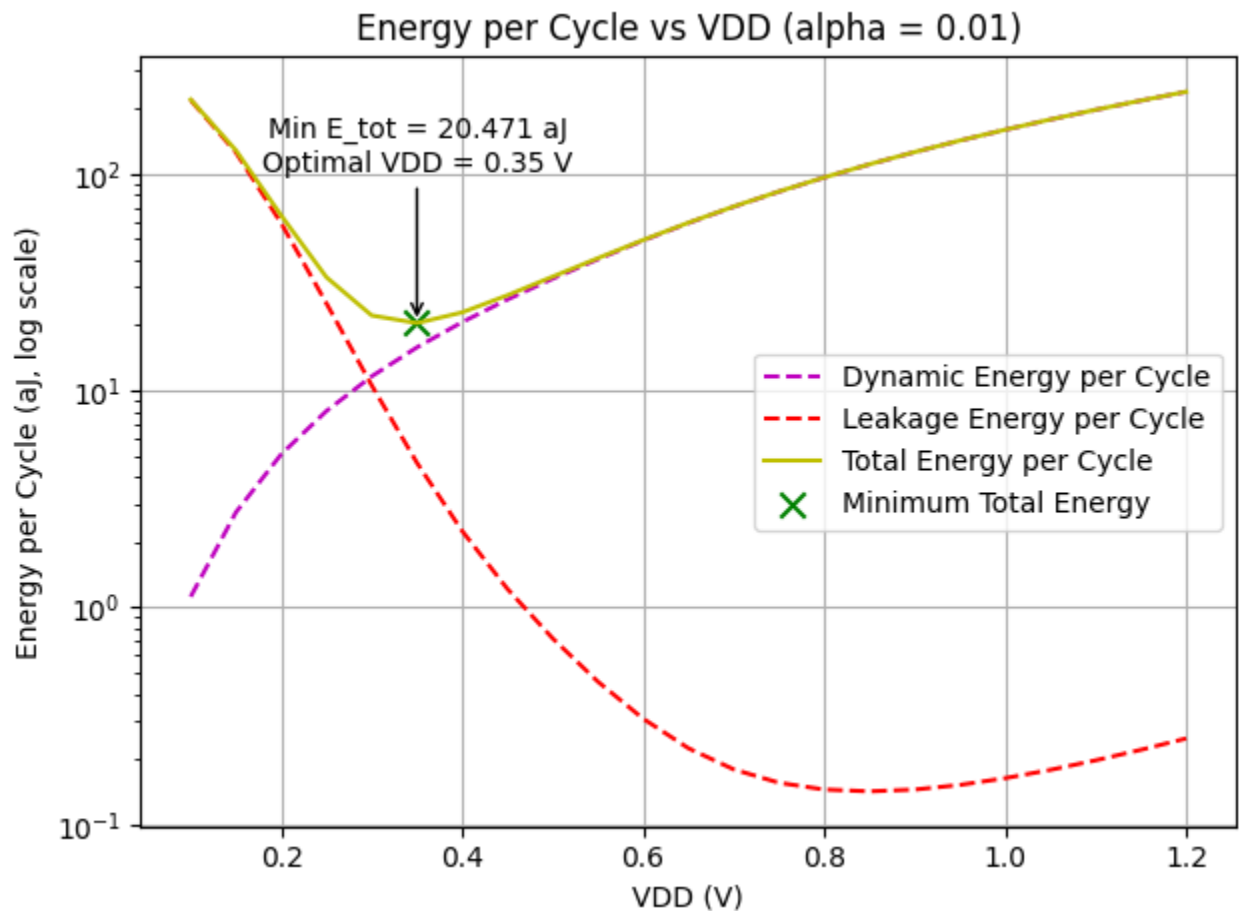
- $P_{dyn} = P_{tot} - P_{leak}$
- $P_{dyn,1\%} = 0.01 \cdot P_{dyn}$
- $E_{dyn} = P_{dyn,1\%} \cdot T$
- $E_{leak} = P_{leak} \cdot T$
- $E_{tot} = E_{dyn} + E_{leak}$

VDD	P_{tot} (W)	P_{leak} (W)	P_{dyn} (W)	$P_{dyn,1\%}$ (W)	E_{dyn} (aJ)	E_{leak} (aJ)	E_{tot} (aJ)
0.10	2.034p	1.344p	690f	6.9f	1.121	218.4	219.5
0.15	8.212p	2.572p	5.64p	56.4f	2.748	125.3	128.1
0.20	41.36p	4.303p	37.06p	370.6f	5.088	59.08	64.17
0.25	218p	6.627p	211.4p	2.114p	8.053	25.25	33.3
0.30	1.071n	9.648p	1.061n	10.61p	11.63	10.57	22.21
0.35	4.611n	13.5p	4.598n	45.98p	15.82	4.647	20.47
0.40	16.85n	18.32p	16.83n	168.3p	20.69	2.252	22.94
0.45	52.51n	24.32p	52.49n	524.9p	26.28	1.218	27.5
0.50	144.4n	31.7p	144.4n	1.444n	32.8	0.7202	33.52
0.55	360.8n	40.7p	360.8n	3.608n	40.33	0.455	40.79
0.60	826.8n	51.63p	826.7n	8.267n	49.19	0.3071	49.5
0.65	1.711u	64.86p	1.711u	17.11n	59.2	0.2244	59.43
0.70	3.173u	80.88p	3.173u	31.72n	70.36	0.1794	70.54
0.75	5.321u	100.2p	5.321u	53.21n	82.64	0.1556	82.79
0.80	8.195u	123.7p	8.195u	81.95n	95.95	0.1449	96.1

0.85	11.8u	151.9p	11.8u	118n	110.3	0.1421	110.4
0.90	16.14u	185.9p	16.14u	161.4n	125.7	0.1448	125.8
0.95	21.19u	226.9p	21.19u	211.9n	142.0	0.1521	142.2
1.00	26.93u	276.2p	26.93u	269.3n	159.3	0.1634	159.5
1.05	33.39u	335.4p	33.39u	333.9n	177.6	0.1784	177.8
1.10	40.54u	406.6p	40.54u	405.4n	196.9	0.1975	197.1
1.15	48.39u	492.1p	48.39u	483.9n	217.1	0.2208	217.3
1.20	56.95u	595p	56.95u	569.5n	238.4	0.2491	238.6

(F) Plot dynamic, leakage and total energy per cycle versus VDD and graphically obtain the value of VDD at which the total energy is minimized. Show the minimum energy and optimal VDD clearly on the plot.

Solution:



(G) Briefly explain in your own words the plot obtained in (F).

Solution:

At very low VDD values, the total energy is high even though dynamic energy is small. This happens because the circuit becomes very slow in the subthreshold region. Since the clock period becomes very large, leakage energy accumulates over a longer time, which makes the total energy per cycle large.

As VDD increases toward the near threshold region, the delay reduces significantly. The clock period becomes much smaller, so leakage energy per cycle drops sharply. At the same time, dynamic energy is still relatively small because VDD is not very high. This creates a minimum point in the total energy curve.

When VDD increases further into the super threshold region, dynamic energy begins to dominate. Since dynamic energy is proportional to V_{DD}^2 , it grows rapidly with VDD. Even though leakage energy continues to decrease slightly due to shorter period, the increase in dynamic energy outweighs this reduction, causing the total energy per cycle to rise again.

As a result, the total energy curve forms a U shape, with the minimum occurring in the near threshold region. In our case, the optimal VDD is around 0.35 V, where the balance between dynamic and leakage energy gives the lowest total energy per cycle.

(H) The results obtained above are for a 13-stage ring oscillator. By assuming the NAND gate power and delay are almost equal to the inverter (although this is not quite true) and using the results of your simulations above, calculate the FO4 delay, dynamic energy per cycle and the leakage power of a single inverter as a function of VDD. Clearly, show how you obtained these results.

Note: The assumption above is reasonable since the total delay/energy/power is the sum of 12 inverters and that of a NAND gate, so the effect of NAND gate is more or less negligible.

Solution:

1) FO4 Delay of a Single Inverter

For an N-stage ring oscillator, the oscillation period is approximately:

$$T = 2N \cdot t_{pd}$$

So the per-stage propagation delay is:

$$t_{pd} = \frac{T}{2N}$$

Since we assume each stage behaves like an inverter, I approximate the FO4 delay as:

$$t_{FO4} = \frac{T}{26}$$

So, for each VDD value, I computed the FO4 delay by dividing the measured oscillation period by 26.

2) Leakage Power of a Single Inverter

From the EN = 0 simulation, I obtained the total leakage power of the entire 13-stage ring.

Since leakage power is approximately additive across stages, I divided the total leakage power by 13 to obtain the leakage power of a single inverter:

$$P_{leak,inv} = \frac{P_{leak}}{13}$$

3) Dynamic Energy per Cycle of a Single Inverter

From the EN = 1 simulation, I obtained the total power of the ring, and from EN = 0 I obtained the leakage power.

The dynamic power of the ring is:

$$P_{dyn} = P_{tot} - P_{leak}$$

The dynamic energy per cycle of the ring is:

$$E_{dyn,ring} = (P_{tot} - P_{leak}) \cdot T$$

Since total energy is approximately the sum of energies from 13 identical stages, I divided this by 13 to obtain the dynamic energy per cycle of a single inverter:

$$E_{dyn,inv} = \frac{(P_{tot} - P_{leak}) \cdot T}{13}$$

I did not include any switching activity factor here because this energy is directly obtained from the oscillating ring measurement. If activity scaling were required, it could be incorporated separately by multiplying activity factor α to the above equation.

Final Results (Single Inverter)

VDD (V)	$t_{FO4}(ns)$	$P_{leak,inv} (pW)$	$E_{dyn,inv}(aJ)$
0.10	6250.0	0.103	8.625
0.15	1874.231	0.198	21.141
0.20	528.077	0.331	39.138
0.25	146.538	0.510	61.949
0.30	42.154	0.742	89.480
0.35	13.238	1.038	121.728
0.40	4.727	1.409	159.124
0.45	1.926	1.871	202.191
0.50	0.874	2.438	252.311
0.55	0.430	3.131	310.253
0.60	0.229	3.972	378.333
0.65	0.133	4.989	455.504
0.70	0.085	6.222	541.349
0.75	0.060	7.708	635.643
0.80	0.045	9.515	738.169
0.85	0.036	11.685	849.044
0.90	0.030	14.300	966.899
0.95	0.026	17.454	1092.577
1.00	0.023	21.246	1225.510
1.05	0.020	25.800	1366.408
1.10	0.019	31.277	1514.622
1.15	0.017	37.854	1670.182
1.20	0.016	45.769	1833.771

One Worked Example at $V_{DD} = 1.1$ V,

$$t_{FO4} = \frac{T}{26}$$

$$P_{leak,inv} = \frac{P_{leak}}{13}$$

$$E_{dyn,inv} = \frac{(P_{tot} - P_{leak}) T}{13}$$

Substituting,

$$t_{FO4} = \frac{0.4857 \text{ ns}}{26} = 0.01868 \text{ ns}$$

$$P_{leak,inv} = \frac{406.6 \text{ pW}}{13} = 31.277 \text{ pW}$$

$$E_{dyn,inv} = \frac{(40.54 \times 10^{-6} \text{ W} - 406.6 \times 10^{-12} \text{ W})(0.4857 \times 10^{-9} \text{ s})}{13} = 1.5146 \times 10^{-15} \text{ J} = 1514.62 \text{ aJ}$$